

Location, location, location: does a tabular foundation model need spatial features?

TabPFN-3.5 against spatial econometrics and gradient boosting on 130,000 declared transaction prices in São Paulo

Agnaldo Calvi Benvenho*

September 28, 2026 — draft

Abstract

Mass appraisal models encode the dependence of property prices on location explicitly, through spatial weights matrices, spatial lags or engineered neighbourhood features. We ask whether a tabular foundation model can do without them. TabPFN-3.5 receives the property attributes and the geographic coordinates as two ordinary numeric columns, with no spatial modelling of any kind, and is compared with an OLS hedonic model, a spatial autoregressive lag model (SAR, GM estimation) and XGBoost with a k -nearest-neighbour spatial lag, rotated coordinates and nested tuning. The data are 82,187 declared transaction prices from the São Paulo real-estate transfer tax (ITBI) in 2025 and 47,810 in the first seven months of 2026, for apartments, houses and commercial units. The evaluation has two legs: leave-one-block-out cross-validation over ten spatial blocks, and an out-of-time test that fits on 2025 and predicts 2026. Refutation criteria were fixed in advance: worse pooled error, losses in most blocks under paired tests, or more residual spatial autocorrelation (Moran's I) than the explicitly spatial baselines. Out of time, the plain-table model beats the spatial XGBoost in ten of ten blocks (RMSE 0.265 vs 0.287 in log unit price, MAPE 19.5% vs 21.0%, block-bootstrap 95% CI of the difference $[-0.032, -0.026]$) and leaves the least residual autocorrelation of the four models ($I = 0.092$). Across unseen blocks it ties the spatial XGBoost on error but leaves more spatial structure in its residuals ($I = 0.348$ vs 0.309), a partial refutation. Permutation SHAP shows the two coordinates as the strongest inputs. **[TODO: Sharpen the last sentence and add one on implications for appraisal practice.]**

Keywords: mass appraisal; tabular foundation models; TabPFN; spatial autoregression; spatial cross-validation; hedonic prices; São Paulo.

1 Introduction

[TODO: Two to three paragraphs. (1) The problem: mass appraisal at city scale, the role of location, and the engineering cost of explicit spatial modelling (weights matrices, lags, GWR-style local models) that must be redone per city and dataset. (2) Tabular foundation models: in-context learning on synthetic priors [5, 6, 12]; the open question of whether spatial dependence is learned from raw coordinates. (3) Contribution and the falsificationist framing: a hypothesis stated with pre-registered refutation criteria and a comparison deliberately asymmetric against it. Cite the reference study on Belo Horizonte listings [3] as the origin of the question.]

The hypothesis is stated so that it can fail:

A tabular foundation model that receives only the property attributes and the coordinates as two plain numeric columns — no spatial weights matrix, no spatial lag, no

*BP Avaliações e Perícias, Santana do Parnaíba, Brazil; IBAPE Nacional. agnaldo.benvenho@gmail.com. Code and data: <https://github.com/abenvenho/tabpfn-itbi-sp>.

engineered neighbourhood features — predicts spatially correlated prices as well as, or better than, specialised models that encode spatial dependence explicitly.

It is refuted if any of the following holds under a shared protocol: worse pooled RMSE or MAPE; losses in most spatial blocks under paired tests; or more spatial autocorrelation left in its out-of-fold residuals than the explicitly spatial models leave in theirs.

Contributions. **[TODO: Bullet or prose: (i) the first comparison of a tabular foundation model with spatial econometric and spatially augmented tree baselines on real declared prices at city scale; (ii) a two-leg protocol with spatial block CV and out-of-time testing and pre-registered refutation criteria; (iii) a public, reproducible pipeline over open municipal data; (iv) the finding itself, including the partial refutation.]**

2 Related work

[TODO: Compact, one page at most. Three threads: spatial hedonic models and their estimation [2, 7, 8]; machine learning for mass appraisal with spatial features, geographically weighted approaches and spatial cross-validation [10, 14]; tabular foundation models and in-context learning [5, 6, 12]. Close with the Belo Horizonte study [3] and what this paper changes: listings $\rightarrow$ declared prices, one type $\rightarrow$ three, 5k $\rightarrow$ 82k rows, added out-of-time leg and residual autocorrelation criterion.]

3 Data

3.1 Source and cleaning

The São Paulo City Hall publishes every paid form of the real-estate transfer tax (ITBI, *Imposto sobre Transmissão de Bens Imóveis*) with the declared transaction price, the cadastral key (SQL: sector, block, lot) and the physical attributes of the property from the annual property-tax register (IPTU): built and lot area, use, construction standard, corrected completion year. The 2025 workbook holds 230,525 forms and the January–July 2026 workbook 135,655. Coordinates come from the centroids of the fiscal-block polygons of the municipal cadastre (GeoSampa), so all properties in one city block share a location; distance to the nearest subway or commuter-rail station is computed from the 203 station points of the same portal.

[TODO: Summarise the filter chain from `data/filter_report.md` in a short table with the record counts after each step (transfer type, use codes, area and price bounds, duplicates, georeferencing).] The cleaned bases hold 82,187 transactions for 2025 (training) and 47,810 for 2026 (test). A stratified sample of 24,000 rows of 2025 (12,000 apartments, 7,000 houses, 5,000 commercial units; “Level A”) is used wherever the slower baselines must run.

3.2 Under-declaration

The ITBI tax base is $\max(\text{declared price}, \text{assessed reference value})$, so a buyer who under-declares has no incentive to declare below the assessed value: under-declared cash deals bunch exactly at it. The pipeline detects that bunching and removes cash transactions declared at the reference value; financed deals are bank-appraised and kept. **[TODO: Two sentences with the counts and a pointer to the diagnostic figure in `data/underdeclaration_report.md`.]** Section 5.3 repeats the whole analysis on financed deals only.

3.3 Descriptive statistics

[TODO: Add a map of the 2026 transactions or reuse Figure 1(a); one paragraph on the spatial structure of prices (Moran’s I of the raw log unit price on the k-NN-8 matrix).]

Table 1: The three bases. Unit value = declared price / built area, in R\$/m².

Base	Rows	Apartments	Houses	Commercial	Financed	Median unit value
2025, full (training)	82,187	60,067	15,940	6,180	34%	4,836
2025, Level A (sample)	24,000	12,000	7,000	5,000	30%	4,715
2026, Jan–Jul (test)	47,810	35,354	9,429	3,027	42%	4,970

4 Methods

4.1 Target and features

The target is $y = \ln(\text{price}/\text{built area})$. The information set is the same for every model: built area, lot area, age, finish grade (IPTU standard level, 0–5), distance to the nearest station, month index, property type, latitude and longitude. The models differ only in what they are allowed to do with the last two.

4.2 Models

OLS hedonic. Log-linear in the areas and the distance, quadratic in age, dummies for type, coordinates as covariates.

SAR lag. $y = \rho W y + X\beta + \varepsilon$ with a row-standardised k -nearest-neighbour weights matrix ($k = 8$) built inside each training fold, estimated by the generalised-moments procedure of Kelejian and Prucha [7] as implemented in `spreg` [13]. Prediction for held-out rows uses the reduced form with neighbours drawn from the training rows only. **[TODO: Confirm the prediction formula used in `src/model_sar.py` and state it.]**

XGBoost with spatial features. The hedonic block above plus UTM coordinates rotated by 30°, 45° and 60° and a k -NN-8 spatial lag of the target computed inside the training fold (leave-one-out for training rows). Hyper-parameters are tuned by nested Optuna (30 trials, TPE), with an inner leave-one-block-out on the training blocks; the outer test block never touches tuning [1, 4]. Three seeds. Geographically weighted XGBoost, used in the reference study, was timed and found infeasible at this scale (one local model per training row and bandwidth candidate over a dense $n \times n$ distance matrix); Appendix B.

TabPFN-3.5, plain table. The nine raw columns, categorical type declared as such, through the hosted model (`tabpf-client` 0.6.0, model version 3.5). Zero-shot, and a thinking-mode variant in which the fit explores configurations by internal validation on the training fold (RMSE objective, medium and high effort). No spatial information of any kind beyond the two coordinate columns. Fitting on 24,000 rows takes about 9 s; on 82,187 rows, 15 s.

4.3 Evaluation protocol

Leg 1: spatial block cross-validation. Ten k -means blocks on the Level A coordinates, each held out in turn [10, 14]. Every fold-internal quantity (weights matrix, spatial lag, tuning) is recomputed on the training blocks only.

Leg 2: out-of-time. One fit on 2025 (Level A, then the full base), predictions for all 47,810 transactions of 2026, one to seven months after the training window. Nothing from 2026 touches fitting or tuning. Leakage check: no transaction (key, date, price) appears in both years; 0.6% of 2026 rows share a cadastral key with Level A.

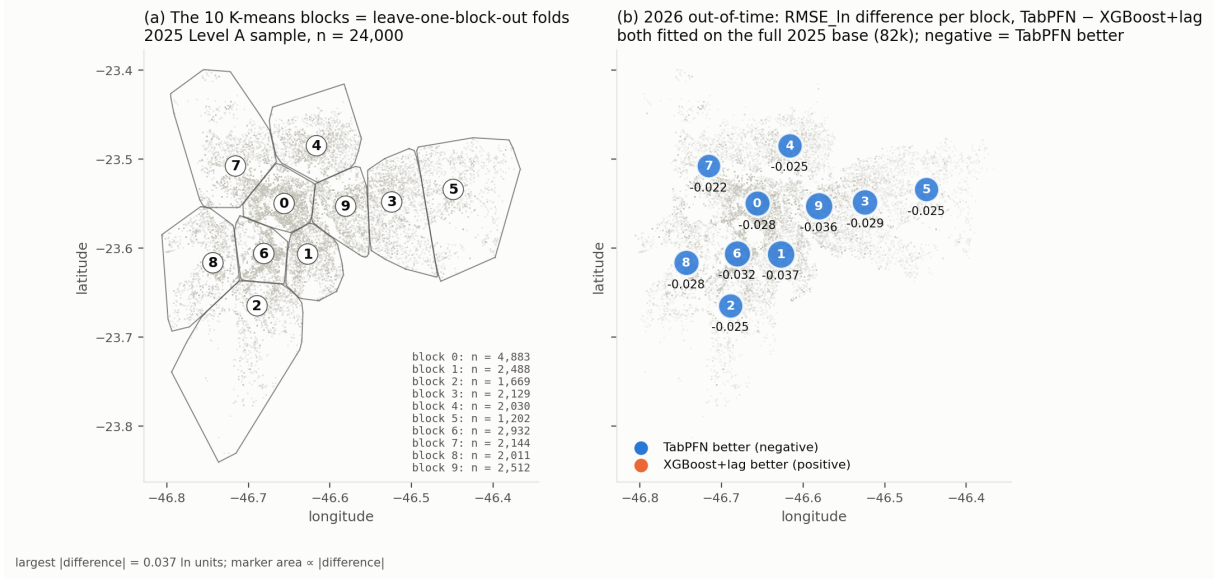


Figure 1: (a) The ten k -means blocks of the Level A sample, each held out in turn. (b) Per-block RMSE difference TabPFN-3.5 – XGBoost+lag on the 2026 test, both fitted on the full 2025 base; negative in every block.

Metrics and tests. RMSE, MAE and R^2 in log units, MAPE in price units, mean residual (bias). Residual spatial autocorrelation by Moran’s I [11] on a k -NN-8 matrix of the test coordinates. Paired comparisons over the ten blocks: Wilcoxon signed-rank test on the per-block metric and a block bootstrap (2,000 resamples of blocks) for the pooled difference with a 95% percentile interval. Where a model is stochastic, three seeds and a 95% interval over seeds.

5 Results

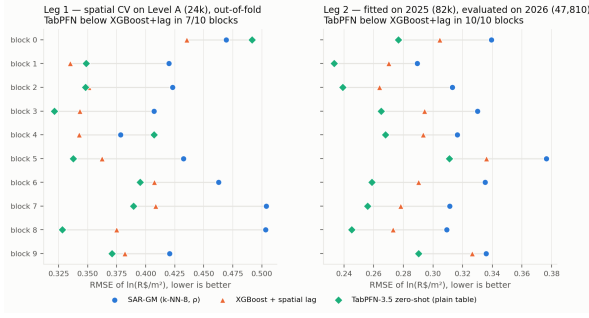
5.1 Leg 1: unseen blocks

Table 2: Leave-one-block-out CV on Level A (24,000 rows of 2025). Out-of-fold metrics in log unit price; Moran’s I of the out-of-fold residuals on a k -NN-8 matrix. Ranges are over three seeds.

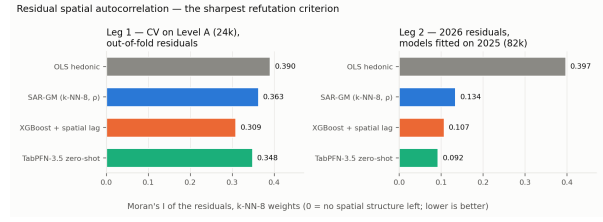
Model	RMSE	MAPE	R^2	Moran’s I
Median by type (floor)	0.497	42.1%	−0.04	0.446
OLS hedonic	0.470	39.4%	0.066	0.390
SAR lag, GM ($\hat{\rho} = 0.91$)	0.448	38.4%	0.155	0.363
XGBoost, rotated coordinates, no lag	0.397	33.3%	0.334	0.355
XGBoost, rotated coordinates + k -NN-8 lag	0.385 (0.384–0.387)	32.3%	0.376	0.309
TabPFN-3.5, plain table, zero-shot	0.394 (0.390–0.394)	32.2%	0.346	0.348
TabPFN-3.5, plain table, thinking (medium)	0.389	31.7%	0.362	0.342
TabPFN-3.5, plain table, thinking (high)	0.388	31.7%	0.366	0.339

TabPFN-3.5 zero-shot beats the SAR lag model in 8 of 10 blocks ($\Delta\text{RMSE} = -0.054$, 95% CI $[-0.100, -0.014]$, Wilcoxon $p = 0.010$; $\Delta\text{MAPE} = -6.2$ pp). Against XGBoost with the explicit lag the difference is not distinguishable from zero ($\Delta\text{RMSE} = +0.009$ $[-0.018, +0.034]$; 7 blocks won, one clear loss in the largest block). Thinking mode improves on zero-shot by a small, consistent margin (-0.005 $[-0.008, -0.001]$); high effort costs about 13 times more per fold and adds nothing beyond noise.

On the residual criterion the hypothesis is partly refuted in this leg: Moran’s I of the plain-table model (0.348) lies below SAR (0.363) but above XGBoost with the explicit lag (0.309).



(a) RMSE per block, both legs.



(b) Residual Moran's I , both legs.

Figure 2: Left of each panel: leave-one-block-out CV on Level A; right: 2026 out-of-time, full 2025 base. **[TODO: Check panel layout against the actual figures and adjust the caption.]**

Within XGBoost, the lag alone is worth $\Delta\text{RMSE} = -0.013$ $[-0.024, -0.004]$ and lowers the residual I from 0.355 to 0.309 — the measurable value of modelling space explicitly when the test is a part of the city no model has seen.

5.2 Leg 2: one year ahead

Table 3: Out-of-time test: fit on 2025, predict the 47,810 transactions of 2026. Bias is the mean residual in log units (positive = under-prediction). Moran's I on a k -NN-8 matrix of the 2026 coordinates.

Training base	Model	RMSE	MAPE	R^2	Bias	Moran's I
Level A (24k)	OLS hedonic	0.404	32.4%	0.185	+0.033	0.408
	SAR lag	0.346	27.0%	0.401	+0.033	0.224
	XGBoost + lag	0.288	21.8%	0.584	+0.047	0.140
	TabPFN-3.5, zero-shot	0.275	20.4%	0.622	+0.055	0.122
	TabPFN-3.5, thinking (medium)	0.275	20.5%	0.621	+0.055	0.121
Full (82k)	OLS hedonic	0.402	32.4%	0.194	+0.023	0.397
	SAR lag	0.326	25.0%	0.470	+0.031	0.134
	XGBoost + lag	0.287 ^a	21.0%	0.59	+0.07–0.10	0.105
	TabPFN-3.5, zero-shot	0.265	19.5%	0.649	+0.058	0.092

^a three seeds: 0.280–0.294.

Paired over the ten 2026 blocks, TabPFN-3.5 zero-shot trained on Level A beats XGBoost with the explicit lag in 10 of 10 ($\Delta\text{RMSE} = -0.013$ $[-0.016, -0.011]$, $\Delta\text{MAPE} = -1.4$ pp, $p = 0.002$) and SAR by -0.071 $[-0.079, -0.062]$. On the full base the gap to XGBoost widens to -0.029 $[-0.032, -0.026]$ ($\Delta\text{MAPE} = -2.0$ pp, 10 of 10) and the plain-table model has the lowest residual autocorrelation of the four. Scaling from 24k to 82k rows improves TabPFN by -0.010 $[-0.012, -0.008]$ and SAR by -0.020 ; XGBoost does not improve ($+0.006$ $[+0.001, +0.011]$) because its trend bias doubles. Errors are flat across blocks (0.25–0.32) and across horizons of one to seven months (0.26–0.29). In this leg the hypothesis survives every criterion, at both scales.

5.3 Robustness: financed transactions only

Cleaner prices lower every error and leave the ranking unchanged. The plain-table model now beats XGBoost+lag in the block CV as well ($\Delta\text{RMSE} = -0.017$ $[-0.032, -0.003]$, $p = 0.049$, 8 of 10 blocks) and the partial refutation on the residual criterion does not reappear: it leaves less spatial structure than the spatial XGBoost in both legs. The headline results do not depend on the cash deals retained by the under-declaration filter.

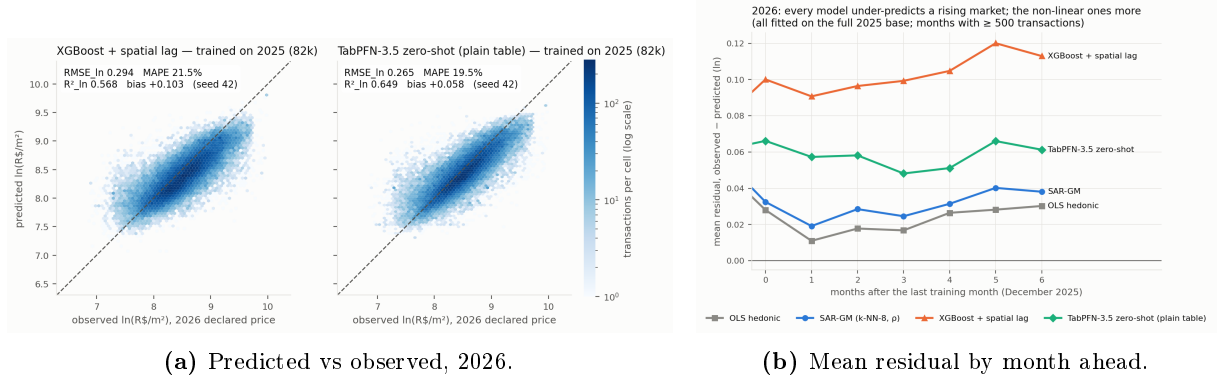


Figure 3: The rising market of 2026: every model under-predicts, the non-linear learners more so.

Table 4: Both legs on financed deals only: 7,211 Level A rows (CV) and the same rows fitted once and scored on the 20,052 financed transactions of 2026.

Model	CV RMSE	CV MAPE	CV I	2026 RMSE	2026 MAPE	2026 I
OLS hedonic	0.354	28.5%	0.343	0.343	26.5%	0.491
SAR lag	0.344	28.1%	0.342	0.286	22.0%	0.297
XGBoost + lag	0.286	23.0%	0.327	0.221	16.7%	0.160
TabPFN-3.5, zero-shot	0.269	21.1%	0.279	0.207	15.3%	0.143

5.4 Explaining an estimate

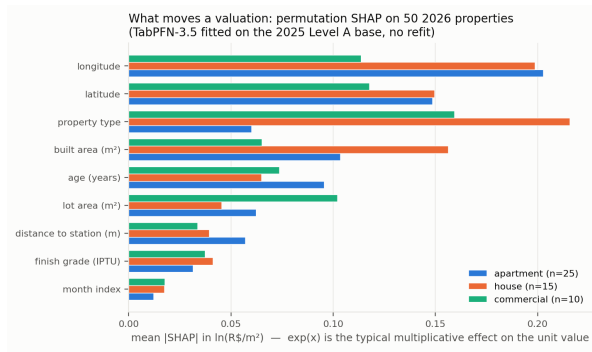
Permutation SHAP [9] on 50 stratified 2026 transactions the model had never seen ranks longitude (mean $|\text{SHAP}|$ 0.184, a typical $\times 1.20$ on the unit value) and latitude (0.143, $\times 1.15$) as the two strongest inputs, ahead of property type (0.127), built area (0.112) and age (0.082). Directions match appraisal experience: age and distance to a station pull the unit value down, finish grade pushes it up, larger built areas lower the price per square metre. The two raw coordinates carry, inside the model, the signal the baselines encode with a weights matrix. **[TODO: One sentence on the reconstructed-prediction check (RMSE 0.296 on the sample).]**

6 Discussion

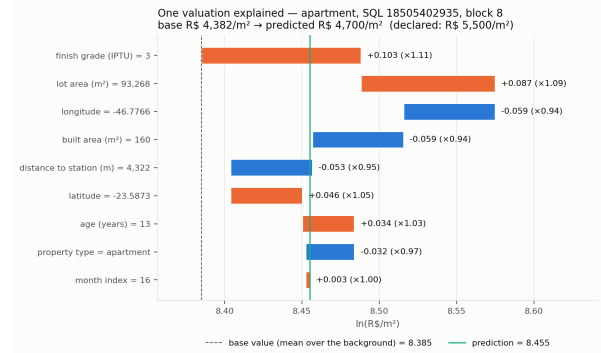
[TODO: Four paragraphs. (1) Reading the two legs together: extrapolation to unseen blocks is where explicit spatial modelling still earns its keep on the residual criterion, while inside the city one year ahead the plain table wins outright; discuss why (in-context learning over 24–82k rows recovers neighbourhood structure from the coordinates when neighbours exist in context). (2) The trend bias: none of the non-linear models extrapolates the month index; what appraisal practice would do (index adjustment) and why it was not applied here. (3) Cost and engineering: seconds against nested tuning and weights matrices; what this means for mass appraisal at municipal scale and for the Brazilian standard NBR 14653-2 (the standard’s precision grades need intervals, which the model provides as predictive quantiles — future work). (4) Limitations: centroid coordinates, declared prices, one city, hosted model (no access to internals, hence no per-row attribution for the regression head), no comparables from inside the model.]

7 Conclusion

[TODO: One paragraph restating the hypothesis, the verdict per leg, and what would change the verdict (another city, listings vs declared prices, a model with row-level attribution).]



(a) Mean |SHAP| by feature and type.



(b) One valuation, base value to estimate.

Figure 4: Permutation SHAP on 50 transactions of 2026 (independent masker, 20-row background), model reloaded from its cached record. Values in log units; e^{SHAP} is the multiplier on the unit value.

Reproducibility

Code, cleaned data, cached model responses and every number in this paper are at <https://github.com/abenvenho/tabpfn-itbi-sp> (Apache-2.0). The pipeline was re-run from a clean clone; the three bases were rebuilt byte-identical from the raw public workbooks.

Acknowledgements

[**TODO:** Prior Labs hackathon; AI assistance statement (code developed with Claude; methodological decisions by the author).]

References

- [1] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631, 2019.
- [2] Luc Anselin. *Spatial Econometrics: Methods and Models*. Kluwer Academic Publishers, Dordrecht, 1988.
- [3] Agnaldo Calvi Benvenho. Métodos não lineares versus regressão espacial na avaliação de apartamentos: um estudo comparativo em Belo Horizonte. *Anais do CEAD 2026*, 2026. CHECK: full congress name, place, pages, award.
- [4] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [5] Noah Hollmann, Samuel Müller, Katharina Eggenberger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations (ICLR)*, 2023.
- [6] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeyer, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637:319–326, 2025. doi: 10.1038/s41586-024-08328-6.

- [7] Harry H. Kelejian and Ingmar R. Prucha. A generalized spatial two-stage least squares procedure for estimating a spatial autoregressive model with autoregressive disturbances. *The Journal of Real Estate Finance and Economics*, 17(1):99–121, 1998.
- [8] James LeSage and R. Kelley Pace. *Introduction to Spatial Econometrics*. Chapman and Hall/CRC, 2009.
- [9] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, pages 4765–4774, 2017.
- [10] Hanna Meyer and Edzer Pebesma. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12(9):1620–1633, 2021.
- [11] P. A. P. Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23, 1950.
- [12] Prior Labs Team. TabPFN-3: Technical report. Technical report, Prior Labs, 2026. arXiv:2605.13986. CHECK: add the TabPFN-3.5 release note or report if one exists.
- [13] Sergio J. Rey and Luc Anselin. PySAL: A Python library of spatial analytical methods. *The Review of Regional Studies*, 37(1):5–27, 2007.
- [14] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

A Filter chain

[**TODO:** Table with the per-step record counts from `data/filter_report.md`.]

B Geographically weighted XGBoost: feasibility pilot

[**TODO:** Two sentences and the timing numbers from `results/gxgb_timing_pilot.json`.]

C Hyper-parameters

[**TODO:** XGBoost search space and the selected values per fold (`results/xgb_params_*.json`); TabPFN-3.5 settings (version, thinking effort, timeout).]